

PRUDHOE BAY DEADHORSE, AK, USA

WMO: 700637

Lat: 70.192N

Lon: 148.477W

Elev: 19

StdP: 101.10

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 27406

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.3	-39.9	-37.7	0.1	-33.5	-37.0	0.1	-33.4	17.6	-21.0	15.8	-20.0	3.3	240	1.068

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	19.7	14.1	17.2	12.8	14.8	11.6	14.5	18.6	13.1	16.6	11.8	14.7	5.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	9.0	16.2	11.0	8.2	15.2	9.7	7.5	14.0	40.6	18.6	37.0	16.8	33.6	14.8	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.0	11.8	DB	-43.2	24.8	2.5	2.0	-45.0	26.3	-46.5	27.5	-47.9	28.6	-49.7	30.1	(4)
(5)			WB	-37.3	16.8	1.6	1.5	-38.4	17.9	-39.4	18.8	-40.3	19.6	-41.5	20.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.0	-25.6	-25.2	-25.2	-15.8	-4.5	4.5	8.7	6.6	1.7	-7.3	-16.4	-22.4
	DBStd	14.15	7.78	8.37	7.89	6.66	5.34	4.34	3.90	4.10	3.72	5.57	6.63	7.30
	HDD10.0	7359	1104	985	1090	775	448	176	73	124	248	537	793	1006
	HDD18.3	10339	1362	1218	1348	1025	707	416	299	363	498	795	1043	1264
	CDD10.0	63	0	0	0	0	0	11	32	19	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	9	0	0	0	0	0	1	7	2	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	5.5	5.8	5.8	5.6	5.5	5.8	5.7	5.2	4.9	5.1	5.5	5.5	5.6
Precipitation	PrecAvg	186	10	7	7	5	7	14	28	37	28	22	12	11
	PrecMax	335	31	23	34	12	22	43	65	89	75	52	32	34
	PrecMin	109	1	0	0	0	1	3	7	10	4	9	2	1
	PrecStd	49	7	5	6	3	5	10	13	20	19	10	8	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.7	-2.2	-2.1	2.2	8.5	21.3	24.1	21.8	16.1	4.0	0.0	-2.0
		MCWB	-1.4	-2.9	-3.1	0.8	6.3	13.7	16.1	15.4	10.9	2.8	-0.4	-2.8
	2%	DB	-6.3	-6.3	-6.3	-1.0	4.8	17.7	20.4	18.3	11.7	1.5	-2.9	-6.6
		MCWB	-6.8	-6.9	-6.9	-1.8	3.2	12.4	14.6	13.5	8.6	0.8	-3.6	-7.0
	5%	DB	-12.0	-9.7	-10.5	-3.7	2.9	14.8	17.8	15.8	9.0	0.3	-5.1	-9.8
		MCWB	-12.5	-10.3	-11.0	-4.4	1.8	11.0	13.4	12.4	7.2	-0.3	-5.6	-10.3
	10%	DB	-15.7	-13.6	-14.6	-6.2	1.7	11.8	15.3	13.2	6.8	-0.8	-7.1	-12.7
		MCWB	-16.1	-14.2	-15.2	-6.7	0.8	9.1	12.2	10.9	5.7	-1.2	-7.3	-13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	-2.5	-2.8	1.0	6.4	14.4	16.7	16.1	11.0	3.1	-0.4	-2.3
		MCDB	-0.2	-2.0	-2.0	2.1	8.3	19.9	22.5	20.9	15.1	3.8	0.0	-1.8
	2%	WB	-6.2	-6.3	-6.4	-1.7	3.2	12.7	15.1	14.1	9.0	0.9	-3.5	-6.7
		MCDB	-5.7	-5.8	-5.9	-1.1	4.6	17.2	19.6	17.2	11.4	1.3	-2.9	-6.4
	5%	WB	-11.4	-9.6	-10.1	-4.3	1.8	11.0	13.7	12.6	7.4	-0.1	-5.6	-9.9
		MCDB	-10.9	-9.0	-9.6	-3.6	2.8	14.6	17.4	15.4	9.1	0.2	-5.1	-9.5
	10%	WB	-15.3	-13.2	-14.3	-6.7	0.8	9.0	12.4	11.0	5.6	-1.2	-7.3	-12.8
		MCDB	-14.9	-12.7	-13.7	-6.3	1.4	11.9	15.4	13.1	6.8	-0.8	-7.0	-12.4
Mean Daily Temperature Range	5% DB	MDBR	6.8	7.1	7.1	7.7	5.0	6.5	7.9	6.3	4.9	5.4	6.7	6.7
		MCDBR	11.9	11.0	9.2	8.5	6.2	12.6	12.7	11.4	10.0	4.8	7.2	9.5
		MCWBR	11.6	10.8	9.0	8.1	5.0	7.9	7.5	6.8	6.9	4.4	7.1	9.3
	5% WB	MCDBR	12.2	11.0	9.3	8.7	5.7	12.3	12.1	10.7	9.4	4.7	7.2	9.6
		MCWBR	11.8	10.8	9.2	8.2	4.7	7.9	7.7	6.8	6.6	4.3	7.1	9.4
Clear-Sky Solar Irradiance	taub		N/A	0.202	0.242	0.298	0.319	0.335	0.347	0.329	0.295	0.218	N/A	N/A
	taud		N/A	2.063	2.220	2.134	2.205	2.350	2.423	2.515	2.525	2.167	N/A	N/A
	Ebn at Noon		0	545	785	828	856	855	830	807	736	507	0	0
	Edh at Noon		0	37	68	105	112	101	92	74	54	32	0	0
All-Sky Solar Radiation	RadAvg		0.00	0.36	1.97	4.33	5.94	6.18	5.09	3.00	1.53	0.47	0.00	0.00
	RadStd		0.00	0.03	0.08	0.19	0.33	0.43	0.40	0.31	0.14	0.06	0.00	0.00

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+1.14	N/A	N/A	N/A	N/A	N/A	-501	-494	N/A	N/A
(47)	Regional (3 neighbors)	+1.14	N/A	N/A	N/A	N/A	N/A	-501	-494	N/A	N/A

Nomenclature: See separate page